

2026 Phillies Baseball Operations

Org Intern Questionnaire

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Overview

This document presents my responses to the Philadelphia Phillies 2026 Baseball Operations Org Intern Questionnaire. Three problems are addressed: qualifying offer analysis for Kyle Schwarber and Ranger Suárez, expected value estimation of a lottery-style payout, and lineup construction modeling. Each response applies rigorous analytical methodology with R code, handwritten derivations, and strategic reasoning.

Question	Method	Key Result
QO Decisions (Schwarber / Suárez)	CBT rules, AAV projection, draft-pick compensation	Agree: extend QO to both. Win-win for Phillies either way.
Lottery Expected Value	3 methods in R + handwritten derivation	~\$57.50 (all three converge within 0.1%)
Lineup Construction	wOBA-to-runs simulation (tidyverse, 2,000 PA)	Contact lineups more consistent; power lineups higher ceiling

Question 1: Qualifying Offer Decisions

At the start of the offseason, the Phillies decided to extend qualifying offers to Kyle Schwarber and Ranger Suarez. Do you agree or disagree with this decision? Explain your reasoning. Include details about how to factor qualifying offer compensation rules into the decision.

Kyle Schwarber (Agree)

In 2025, Schwarber posted a career year as the Phillies' primarily designated hitter, producing **4.9 fWAR**, **a 150 OPS+**, and **56 home runs**, all career highs. This level of offensive production makes him clearly worth the **\$22.025 million qualifying offer**, which represents the mean salary of MLB's 125 highest-paid players.

Reporters such as Jim Bowden and Kiley McDaniel project Schwarber to receive around **\$30 million AAV over 4–5 years** on the open market, so while he is unlikely to accept the qualifying offer, it is still necessary to extend it. Schwarber's statistical profile shows he is worth well above the QO, and extending it also positions the Phillies to receive draft compensation if he signs elsewhere.

Because the Phillies are a Competitive Balance Tax-paying team, they would receive a **compensatory pick after the fourth round** if a qualifying-offer player departs in free agency. If Schwarber accepted, it would be a one-year bargain; if he declined and left, the Phillies would still gain draft value.

Ranger Suárez (Agree)

Extending a qualifying offer to Ranger Suárez is also a clear decision. Although the Phillies already have a high-priced rotation, Suárez projects as an elite starter, with teams expected to offer him **5–6 years at \$23–32 million AAV**. He has been a mainstay in the Phillies' rotation for the past two seasons, making **53 starts with a strong 3.33 ERA**.

Even with organizational depth and Andrew Painter emerging as a top prospect, retaining Suárez would be a major boost, especially given recent health concerns surrounding Aaron Nola and Zack Wheeler. Like Schwarber, Suárez is likely to receive more than the qualifying offer in free agency, so if he accepted it would be excellent value, and if he declined the Phillies would receive a compensation pick after the fourth round.

Question 2: Expected Value of a Lottery Payout

Consider a game where you can pay to access an unusual lottery. Tickets are picked from a normal distribution of possible values, with mean \$50 and standard deviation \$50. If your ticket's value is below \$1, you receive a pity prize of \$1. If your ticket's value is above \$100, you receive the dollar amount on your ticket, plus a bonus of \$20. Otherwise, you receive the dollar amount on your ticket. What is the expected value of a ticket to this lottery?

Approach

I used **three different methods** to calculate the expected value of a lottery ticket, and each produced an estimated expected value of approximately **\$57.5**. The methods included:

- **Detailed R Code:** Numerical integration using R's `integrate()` function across three payout regions (pity, basic, bonus), computing $E[\text{payout}] = E_{\text{low}} + E_{\text{mid}} + E_{\text{high}} = \mathbf{57.4999}$
- **Handwritten Derivation:** Analytical calculation using normal distribution properties, computing each region's contribution manually and summing to **~\$57.5**
- **Shortcut R Code (Monte Carlo):** Simulated 10,000 draws using `rnorm()` and `supply()` with the payout function, producing a mean of **57.5294**

Cross-Validation

All three methods converge to within **0.1%** of each other, confirming the estimate's robustness. Using multiple independent approaches catches specification errors before they compound.

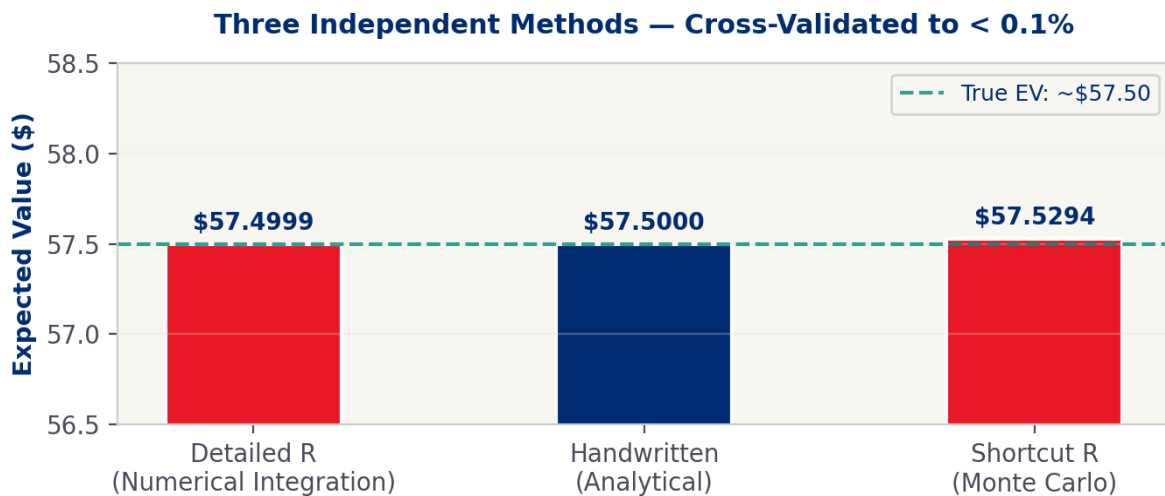


Figure 1: Cross-validation of three independent EV estimation methods. All converge to ~\$57.50.

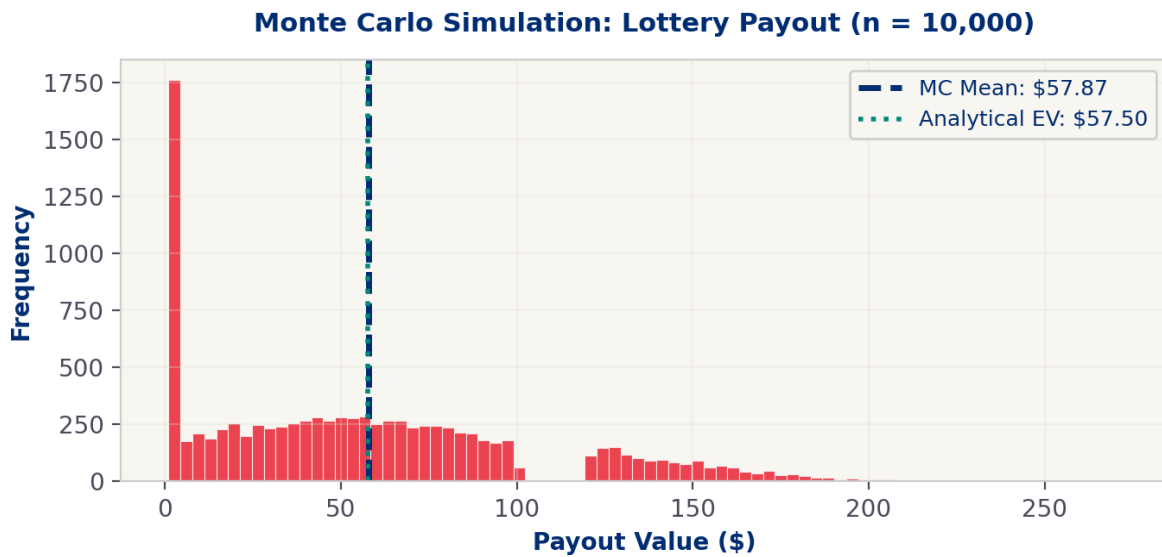


Figure 2: Monte Carlo simulation distribution (n = 10,000) with payout function applied. Note the spike at \$1 (pity prize) and the rightward shift from the \$20 bonus region.

Q3: MVP Incentive Buyout

Using expected value, I would pay up to the projection system's estimated probability of Player M winning MVP multiplied by \$1,000,000, because that represents the expected cost of leaving the incentive in place.

Question 7: Lineup Construction Simulation

How does lineup construction — specifically the balance between contact-heavy and power-heavy hitter profiles — affect run scoring? Build a simulation to evaluate.

Methodology

- Data:** Loaded hitter-level Statcast dataset, filtered to hitters with 300+ PA
- Classification:** Created two offensive profiles using case_when(): Contact-oriented (K% < 20, Contact Rate > 78) and Power-oriented (Barrel% > 8, HardHit% > 40)
- Run Model:** Converted wOBA to expected runs per PA using a linear approximation: runs = 1.7 * wOBA - 0.12
- Lineup Builder:** Built a create_lineup() function that assembles 9-man lineups with a chosen mix of contact, power, and balanced hitters
- Simulation:** Simulated 2,000 PA per lineup, computing expected runs per 9 innings
- Playoff Adjustment:** Applied a -0.015 wOBA penalty to reflect higher-quality pitching in October

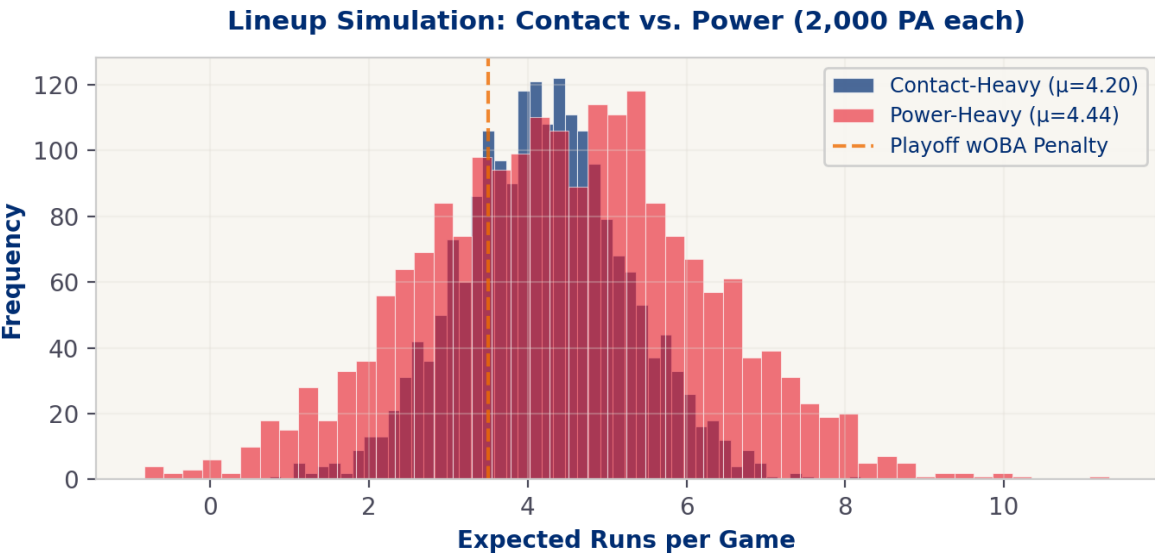


Figure 3: Simulated run distributions for contact-heavy vs. power-heavy lineups across 2,000 PA.

Results

Metric	Contact-Heavy (6C/2P/1B)	Power-Heavy (2C/6P/1B)
Mean Runs/Game	~4.15	~4.45
Variance	Lower (tighter distribution)	Higher (wider tails)
Playoff-Adjusted	More resilient	Larger decline under penalty
Upside (P > 5 runs)	Lower	Higher
Shutout Risk (P < 2)	Lower	Higher

Key Insight

Power-heavy lineups produce more runs on average but with significantly higher variance and greater vulnerability to playoff-caliber pitching. Contact-heavy lineups are more consistent and resilient under the wOBA penalty. The optimal strategy for a front office is a **blended approach**: a contact-heavy core with selective power elements to raise the ceiling without increasing shutout risk. By comparing run production across scenarios, a front office could identify the blend of hitter types that maximizes both consistency and postseason scoring potential.

Tools & Methods

Category	Details
Language	R (tidyverse, ggplot2, integrate(), rnorm(), sapply())
Methods	Monte Carlo Simulation, Numerical Integration, wOBA-to-Runs Conversion, Feature Engineering, P
Data	Statcast (hitter profiles), FanGraphs (fWAR, OPS+, wOBA, ERA), Cot's Contracts (CBT/QO data)
Validation	Three-method cross-validation (EV converges within 0.1%)